

A Net Cure or Curse?

Tracking the Impact of E-Commerce on Urban Freight Transport Intensity in Bogotá and Buenos Aires

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Abstract

The growth of e-commerce has the potential to reduce shopping-related travel but brings with it additional freight vehicle trips for the delivery of online orders to consumers. Understanding the overall net effect of e-commerce on urban trip intensity is essential for planning transport infrastructure and services. The paper analyzes how the growth of e-commerce is impacting mobility in Bogotá and Buenos Aires. The demand for e-commerce grew in both cities during the COVID-19 pandemic (2019–21),

mostly among higher income groups. Despite the significant potential for replacing private vehicle trips, the analysis finds little evidence that the growth of e-commerce is having a significant substitution effect on shopping trips. Overall, e-commerce currently generates more traffic than it avoids in both Bogotá and Buenos Aires, and, thus, is very likely to continue to add to the road traffic in the two metropolitan areas in the near future.

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A Net Cure or Curse? Tracking the Impact of E-Commerce on Urban Freight Transport Intensity in Bogotá and Buenos Aires

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1. Introduction

E-commerce is not a phenomenon that Colombians or Argentines only came to know during the pandemic. In 2019, 136 million annual transactions related to e-commerce were recorded in Colombia (Blacksip 2022), or an average of 2.7 transactions per person. The growth of e-commerce in Colombia was very high in the years prior to the pandemic, rising from 17 million transactions in 2014 to 136 million in 2019. The growth rate was higher than in the rest of the large countries in the Latin America region. In the case of Argentina, 89 million e-commerce purchase orders were made in 2019 (CACE 2020), or around 2 purchase orders per person. The growth of e-commerce in Argentina before the pandemic went through three phases: (i) an eruption in 2000-2005, with growth rates in the number of users and sales exceeding 50 percent; (ii) a consolidation in 2005-2015, during which the share of Argentines connected to the Internet who had tried e-commerce rose from 15 to 75 percent and year-on-year growth rates in sales adjusted for inflation remained above 25 percent; and (iii) a phase of relative stagnation in the years prior to the pandemic, probably due to limited economic growth.

Bogota and Buenos Aires are the most active e-commerce demand centers in Colombia and Argentina, respectively. In the case of Bogota, data reported by marketplaces such as Rappi show a share close to 60 percent in 2019 (Blacksip 2020), and Colombia's Quality of Life Survey (LCS) suggests that Bogota's e-commerce adoption is double the national average. The highest volume of online orders in 2019 was concentrated in the northeastern localities of the city (i.e., Usaquén, Chapinero, and Teusaquillo), coinciding with a greater adoption of e-commerce among households of higher socioeconomic status. In comparison, while in 2010 almost 70 percent of Argentina's online purchases were made in Buenos Aires and its metropolitan area (AMBA for its Spanish acronym), this share declined to around 40 percent in recent years, as other areas of Argentina increased e-commerce adoption, thus decreasing AMBA's weight. Nevertheless, AMBA still concentrates a large part of e-commerce transactions, especially among everyday shoppers. The share of the City of Buenos Aires (CABA) households using the Internet who buy or sell products online rose from 16.9 percent to 31.4 percent in 2011-2018, while in the Province of Buenos Aires the increase was even greater in relative terms, almost tripling from 5.9 percent to 15.9 percent, according to the National Institute of Statistics and Census of Argentina (INDEC). Similarly to Bogota, in Buenos Aires e-commerce adoption is higher among higher-income households, with the gap in adoption rates between the highest and the lowest income groups widening over time. In 2011-2018, the e-commerce adoption rate among the lowest income decile households in CABA increased from nearly none to 12.5 percent, compared to an increase from 15 to 52 percent among the highest income decile.

COVID-19 introduced changes in travel patterns and disruptions in urban mobility, including due to a rise in e-commerce related trips. The mobility restriction and social distancing measures imposed during the COVID-19 pandemic resulted in an increase of e-commerce at a global level. The growth of e-commerce may also change shopping-related travel demand patterns in the long run and generate additional freight vehicle trips for the delivery of online orders to consumers, which urban mobility planning should take into account. The current study aimed to leverage innovative data tools to analyze the growth of e-commerce during the pandemic in Bogota and Buenos Aires and its impacts on trip volumes. The specific objectives were: to characterize e-commerce demand in the two metropolitan areas and its evolution during the COVID-19 pandemic; to quantify the potential reduction in travel demand due to e-commerce; to quantify the weight of e-commerce in urban goods distribution flows; to identify whether the growth of e-commerce during the pandemic has resulted in a reduction in travel demand; and to identify whether the growth of e-commerce during the pandemic has resulted in an increase in urban goods distribution flows.

2. Insights from the literature

E-commerce, or electronic commerce, is the sale or purchase of goods or services, carried out by means of computer networks through methods specifically designed for the purpose of receiving or processing orders, regardless of whether payment and delivery of the goods or services occur online, as defined by OECD (2011). In most cases, e-commerce is not an activity that occurs entirely in an online environment. As with other processes of virtualization of human activity made possible by information communication technologies (e.g., teleworking, telemedicine, etc.), e-commerce has led to a "fragmentation of commercial activity" (Couclelis 2004), since it separates in space and time the sub-processes that are part of the purchase and sale of goods or services.

Depending on the profile of the buyer and seller, the main types of e-commerce include (i) business-to-business (B2B), or electronic commerce of goods and services between companies; (ii) business-to-consumer (B2C), or the purchase of goods and services by individual consumers from companies via the Internet; (iii) consumer-to-consumer (C2C), or the sale and purchase of goods and services between private individuals over the Internet – usually associated with the so-called sharing economy platforms; and (iv) consumer-to-business (C2B), or the electronic purchase of goods and services by companies from individuals, such as the advertising of companies on blogs or social networks of individuals or the search for individuals for online panel surveys. Among these, B2B and B2C represent the main types of goods distribution flows (Cataruzza et al. 2017). With regard to the distribution mechanisms of the products purchased online, which are of particular interest for tracking the impact of e-commerce on mobility, the options include direct distribution (home delivery) and several types of indirect distribution, such as pickup at a physical establishment of the supplier (click and collect), pickup at another establishment, and collection at a box office (Zhang et al. 2019).

Studies to date show that the growth of B2B and B2C e-commerce implies a growth in urban distribution flows of goods (Visser et al. 2014; Pettersson et al. 2018), since there is a greater fragmentation of goods to reach each consumer (replacing large flows to retail outlets), the frequency of purchase grows, and the admissible delivery time for the buyer decreases (Macharis and Kin 2017). Another reason is the higher percentage of failed deliveries in B2C/C2C e-commerce, which have to be repeated, generating more displacement of goods (Bjørgen and Ryghaug 2022). In addition, B2C/C2C e-commerce is considered to be accompanied by a greater spatial dispersion of flows, since, instead of being concentrated between production sites and specific points of sale, part of them are now destined for residential areas, leading to noticeable growth in light-duty vehicles within cities, especially in residential areas (Visser et al. 2014; Bjørgen and Ryghaug 2022). This has important implications on the consumption of public space for parking, congestion resulting from increased traffic flow and loading and unloading operations, as well as on noise and air pollution associated with road traffic.

Although a majority of the population is able to consume products through e-commerce, the adoption of e-commerce is not equal among all population groups. The sociodemographic factors most highlighted by the literature on e-commerce adoption are very similar to those of other ICT-based innovations and include age, income, education, and internet use (Shi et al. 2019). Younger people are more likely to shop online, a trend also confirmed in Colombia (República 2021). Higher e-commerce adoption is also distinctly associated with income: in addition to the income available to purchase goods and services, higher income usually means more Internet access and availability of bank accounts – conditions necessary to access e-commerce. In the case of Colombia, socioeconomic status has been found to be a very relevant factor in explaining e-commerce adoption (Sánchez-Torres et al. 2017), with the higher socioeconomic strata (4 to 6) having the highest adoption (República 2021). Studies focused on Colombia also confirm that e-commerce adoption increases with education (Sánchez-Torres, et al. 2017). The usability of platforms appears as a prominent factor in the propensity to shop online according to a study focused on young people in Medellín, Colombia (Chamorro, et al. 2019). Finally, higher population density of the person's place of residence (e.g., urban areas) is associated with greater adoption of e-commerce, explained by higher Internet penetration, among other factors.

The potential impacts of e-commerce on the mobility of people are usually classified into four types of effects: *substitution*, or a reduction in travel demand due to the virtualization of activities that motivated travel; *complementarity*, or an increase in travel demand because the virtualization of the activity allows or promotes other activities that do require travel; *modification*, or changes in demand patterns not directly related to an increase or decrease in the number of trips; and *neutrality*, or the absence of impact on travel demand. Studies to date have found both substitution and complementarity effects, with the former being slightly more common (Le et al. 2022). Studies that find complementarity effects suggest that the possibility of doing more information search online (one of the activities of e-commerce) not only promotes more online shopping but also more physical shopping (Xi et al. 2020). The studies that find substitution effects have indicated some factors that may enhance this effect, such as shoppers not having their own vehicle (Shi et al. 2019) and same-day deliveries (Xi et al. 2020). Ferrell (2005) quantified the substitution relationship with data from California, finding that for every 100 minutes of time spent on e-commerce, consumers reduced 5 minutes of time spent on shopping trips. There is no consensus among studies that incorporate other dimensions of travel demand such as trip distance, trip chaining, mode choice, or trip frequency for other purposes. Modification effects have been detected in the use cases addressed in each study, but often with contradictory results from each other (Le et al. 2022).

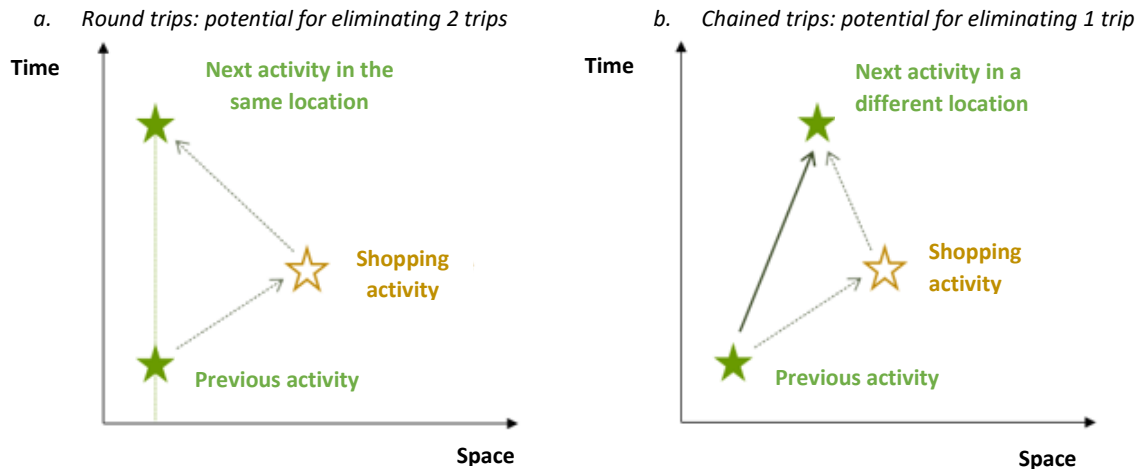
3. Methodology

The current paper analyzed how the growth of e-commerce is impacting mobility in Bogotá and Buenos Aires and how the patterns in e-commerce changed during the pandemic. A distinction can be made between two types of impacts of e-commerce on mobility: (i) its effects on the mobility of people, insofar as they allow the virtual development of the activity of buying products, which required physical interaction before the arrival of e-commerce that generates displacements; and (ii) its effects on the urban distribution of goods, insofar as they can modify the patterns of generation and spatial and temporal distribution of trips in vehicles destined for the distribution of goods between sellers and consumers. Specifically, the analysis: (i) characterizes the demand for e-commerce in both cities and its evolution during the pandemic, (ii) quantifies the potential reduction in travel demand due to e-commerce, (iii) quantifies the share of e-commerce in urban distribution flows of goods, and (iv) identifies whether the growth of e-commerce has resulted in a reduction in travel demand or, rather an increase in urban distribution flows of goods. The weight of trips for shopping purposes in overall mobility and their composition were analyzed according to the following criteria:

- **Home-based or non-home-based travel:** This analysis allows identifying the share of trips that correspond to shopping trips for each of the demand segments of the origin-destination matrices from cell phone data.
- **Type of trip chain.** Two types were identified according to the sequencing of the shopping activity with other activities of the same traveler: (i) round-trip, in which the traveler returns after the shopping activity to the same place where the previous activity took place; and (ii) chained, in which the traveler adds a shopping activity in a trip between two other activities. This analysis allows identifying the potential for trip reduction as a consequence of the substitution of shipping trips by e-commerce, since this substitution can avoid the two trips of the round-trip chains, but only one of the trips of the chained chains (Figure 1).

With different types of agents and multiple impacts on mobility, there are a large number of data sources that can contribute to understanding the demand for e-commerce and its impacts on mobility. A distinction can be made between two main groups of data sources: (i) e-commerce sector data sources, which are those that can characterize indicators related to the adoption and use of e-commerce among consumers, spatial patterns of order demand and the properties of the logistics chains that support e-commerce (Buldeo Rai and Dablan 2022); and (ii) mobility data sources, which are those that can be used to directly measure the impacts of e-commerce on personal travel demand and urban distribution of goods.

Figure 1: Conceptual scheme of the types of trip chains considered



Telecommunications and digital technology provide an opportunity to measure mobility like never before. In data poor settings with low smartphone penetration, call detail records (CDR) have the highest coverage. In Argentina, there were over 56.3 million cellular subscriptions registered in 2019,² and smartphone adoption is expected to increase from 59 percent in 2019 to 82 percent by 2025, while in Colombia the respective figures are 65 percent and 77 percent. The analysis took advantage of the very high mobile phone penetration rates in Argentina and Colombia and the associated availability of mobile phone based, high frequency and spatial resolution mobility data. The analysis is based primarily on the origin-destination matrices generated in this project from cell phone data, or CDR, allowing for a longitudinal analysis of travel demand to shopping areas and carrier movements from e-commerce distribution centers. The anonymized cell phone data was obtained from mobile phone service operators in Argentina and Colombia that represent 37 percent and 20 percent of the respective national markets, each with 8 million to 9 million unique subscribers. CDR-based OD matrices for each of the two cities were constructed for October 2019, 2020, and 2021, allowing to explore the impact of e-commerce growth on travel demand in both cities during the pandemic. The matrices are referenced to the detail zoning employed by the two cities' strategic transport models, cover three day types (weekday, Saturday and Sunday) and are segmented by trip residence location, trip type (passenger vs. professional driver), trip purpose, mode of transport, and trip distance.³

The mobility indicators are calculated by taking as total trips those identified as personal mobility, filtering out from the OD matrices those associated with professional mobility. The methodology for identifying professional mobility and overall methodology for calculating OD matrices generally follows the methodology that leverages CDR data for mobility analysis (e.g., Alexander et al. 2015, Bachir et al. 2019, Bayir et al. 2010, Huang et al. 2019). The methodology develops origin-destination matrices based on anonymized mobile phone data. The origin-destination matrices have limitations, and it is difficult to separately identify trips for shopping purposes. The analysis of cell phone data makes it possible to differentiate some trip purposes by longitudinal analysis of the users' activities in the sample (e.g., trips to home, by identifying the place of residence, or trips to work or study, by identifying the place other than the residence where a long-term recurrent activity takes place). In the case of the origin-destination matrices obtained in the present project, four types of purposes are distinguished: (i) NHB (non-home-based), (ii) HBW (home-based with full-time work purpose), (iii) HBEdU (home-based with study purpose) and (iv) HBO (home-based with purpose other than HBW or HBEdU). Therefore, trips for shopping purposes are spread between the HBO and the NHB categories. The analysis of the mobility survey (Phase 2 of the methodology) provides information about this distribution.

² World Bank World Development Indicators.

³ Matrices are also disaggregated by gender for Buenos Aires; the results are presented in a separate paper.

This limitation is addressed by applying an approach similar to the difference-in-differences method widely used to identify how the evolution of a given part of a system differs from the rest of the system if “treated” by an intervention (Angrist and Krueger 1999). In this case, the application of the method is aimed at identifying whether the typical shopping-purpose trip attractor zones within the urban area have had a different evolution of mobility demand for the HBO and NHB groups than other parts of the urban area. The origin-destination matrices identify trips by carriers (cab drivers, public transport drivers, urban goods distribution vehicles, etc.) in a grouped manner. Specifically, two groups of zones are defined:

- “Treatment” group, or areas that are considered to be shopping trip attractor zones and that are likely to have been impacted by the potential substitution effect created by the growth of e-commerce (the “treatment”, in this context), such as areas in which a major shopping center is located, areas in which a major commercial center and adjacent areas are located, and areas that concentrate one-third of trips destined for shopping activities, according to the 2019 household mobility survey; and
- Comparison group, which includes other areas of the city.

A series of outcome metrics are defined and their evolution over time is calculated within the “treatment” and the comparison areas to subsequently compare the two, including, among others: number of trips during the morning (9:00-12:00) and the evening (19:00-20:00) shopping peak hours; and travel from areas inhabited by different income groups⁴ to the two types of areas, in particular, focusing on the higher income groups that are known to have higher e-commerce adoption rates.

The CDR-based indicators were combined with complementary data, including, among others:

- For Bogota: E-commerce sales reports for Colombia; a tool developed by the Latin American Center for Logistics Innovation (CLI) that integrates order shipment data from three e-commerce companies in 2019; National Quality of Life Survey (ECV); Bogotá Household Mobility Survey 2019; and SDM Bogotá (2021) which includes interviews with 2,126 load generating and attracting establishments in Bogota in 2020-2021 and classified cargo vehicle traffic counts in Bogota in 2015 and October 2020.
- For Buenos Aires: E-commerce reports for Argentina produced by the Argentine Chamber of Electronic Commerce; e-commerce survey of AMBA consumers in 2020; National Household Expenditure Survey (2017-2018); classified traffic counts of freight vehicles in Buenos Aires in 2018-2019 collected by the Undersecretary of Mobility Planning of CABA; and the ENMODO household travel surveys for Buenos Aires (2008-2009, 2018-2020⁵).

4. E-commerce-related mobility prior to the pandemic and potential for shopping trip elimination

Trips linked to shopping activities in 2019 accounted for about 10.5 percent of total mobility in Bogotá on weekdays (EDM Bogota 2019). About 85 percent of shopping trips in 2019 were reported as home-based, i.e., originating from or destined for home. However, it should be noted that household surveys tend to underrepresent non-home-based trips and trips made for activities of short duration (Chapleau 2018) and overestimate the proportion of home-based trips due to response bias (Aschauer et al. 2018). Nearly 10 percent of all home-based trips, one-fifth of the home-based trips other than “compulsory mobility” (work, education), and 15 percent of non-home-based trips were for shopping purposes. About 25 percent of shopping trips in Bogota in 2019 were part of a trip chain, the percentage being much higher among non-home-

⁴ Income groups are approximated by the spatial classification of socioeconomic wellbeing used by the city planning agencies: in Bogota, according to the “strata” classification of city blocks, where 1 (low-low), 2 (low), 3 (low-medium), 4 (medium), 5 (medium-high), and 6 (high); in Buenos Aires, according to the share of unsatisfied basic needs (Necesidades Básicas Insatisfechas, or NBI) of the population inhabiting a city block (deciles) available from the National Institute of Statistics and Census of Argentina (INDEC) at the census radius level for most of the study area.

⁵ It includes a complete characterization of the trips in its sample, in particular, trips for shopping purposes.

based trips (86 percent) than among home-based trips (14 percent), while three-quarters were round trips. The values are similar for the metropolitan region as a whole and for Bogotá.

The INDEC survey data, which indicates an increase in e-commerce adoption in 2011-2018 in CABA (from 16.9 percent to 31.4 percent) and the province of Buenos Aires (from 5.9 percent to 15.9 percent), allows identifying whether this growth translated into a reduction in the number of trips for shopping purposes. In 2018, shopping trips in AMBA accounted for 13 percent of all trips and 12.1 percent of all home-based trips. About 96 percent of all shopping trips in 2018 were home-based, which is significantly higher than in Bogotá; however, this figure is again subject to the caveat that the share home-based trips in surveys tend to be overestimated. Shopping trip generation rates increased between 2011 and 2018, from 1.74 to 1.84 trips per person, suggesting that at the aggregate level there was no trip substitution (reduction) due to the growth of e-commerce.

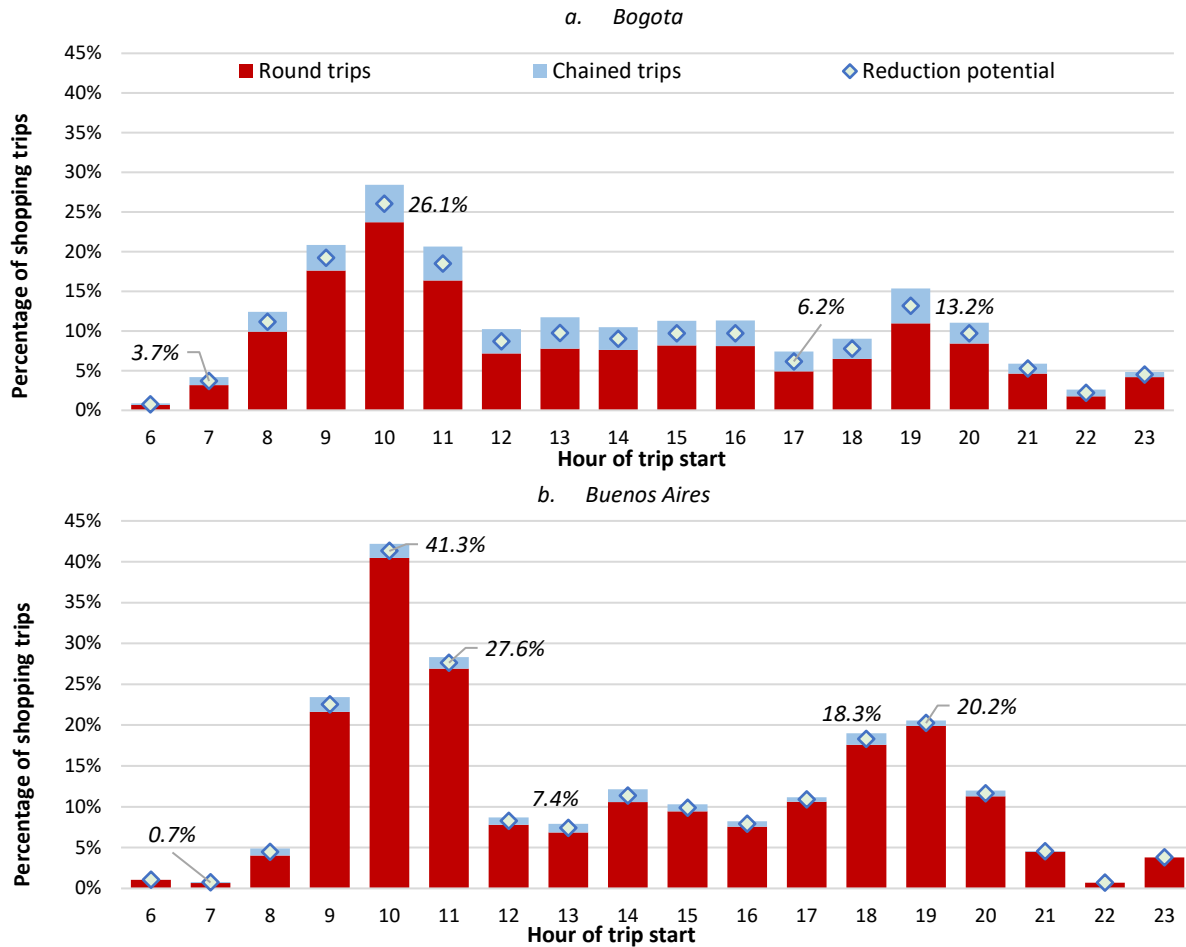
As already noted, the potential for trip reduction by e-commerce depends on the proportion of trips for shopping purposes that are part of trip chains. If people make trips to retail outlets to and from their homes, a virtualization of shopping would lead to a reduction of two trips. Conversely, if people shop as an intermediate activity between work and home, the shift from physical shopping to the Internet would lead to a reduction of one trip instead of two trips. Moreover, if the retail outlet is already located on the route between work and home and the trip is made by car, the impact of eliminating the intermediate activity on road traffic and its externalities will be small (the number of kilometers traveled is maintained).

The total trip reduction potential as a result of e-commerce in Bogotá in 2019 on a typical working day was 9.1 percent, since a total virtualization of commercial activity would lead to a 100 percent reduction of the round trips with a shopping purpose (7.9 percent of trips) but only one of the two trips with origin or destination to shopping activities in the case of trip chains (modifying the origin or destination of the other trips in the chain, avoiding the stop at the commercial establishment), which could lead to a 1.2 percent reduction of trips. In Buenos Aires, about 7 percent of trips with a shopping purpose were part of a trip chain in 2018, the share being much higher among non-home-based trips (93 percent) than among home-based trips (3.7 percent). Thus, the total trip reduction potential due to e-commerce in Buenos Aires on weekdays was 11.7 percent (assuming a full elimination of round-trip trips to or from a shopping activity and elimination of one of the two trips with origin or destination to shopping activities in trip chains).

Trips for shopping purposes in both cities have a somewhat higher weight in the higher income groups, but with a higher proportion of linked trips (especially in Bogotá), which makes the potential for mobility reduction similar for all groups. In Bogotá, the percentage of total trips for shopping purpose peaks in stratum 6 (11.7 percent), but so does the proportion of linked trips out of total trips for this purpose (41.6 percent, compared to 24 percent for the population as a whole). Stratum 2, which is the most populated, has the lowest mobility reduction potential (8.8 percent). In Buenos Aires, trips for shopping purposes in 2018 had a relatively homogeneous share across all income groups although with a positive correlation. The potential for trip reduction in 2018 ranged from 10.9 percent among the lowest income quintile to 12.4 percent among the highest quintile.

The potential for reduction in inter-peak hours is considerable but at peak hours is very small in either Bogotá or Buenos Aires, especially in the morning peak hour (in Bogotá, 3.7 percent, compared to 9.1 percent for the day as a whole, and in Buenos Aires, 0.7 percent, compared to 11.7 in the day overall). Trips for shopping purposes reach the highest weight in overall mobility in the morning at 10:00 (after the morning peak hours), at 28 percent in Bogotá and 42 percent in Buenos Aires (Figure 2). At this time, up to 26.1 percent of trips in Bogotá and up to 41.3 percent in Buenos Aires could be avoided by e-commerce according to the analysis of the household mobility surveys of 2019 and 2018, respectively. In the late afternoon, a new peak is recorded in both cities, but it is about half the intensity as in the morning. The share of linked trips (part of trip chains) in Bogotá is much higher in the evening hours than in the morning, but this is not the case in Buenos Aires.

Figure 2: Share of shopping trips in Bogota and Buenos Aires, by type of travel chain and trip start time (%)

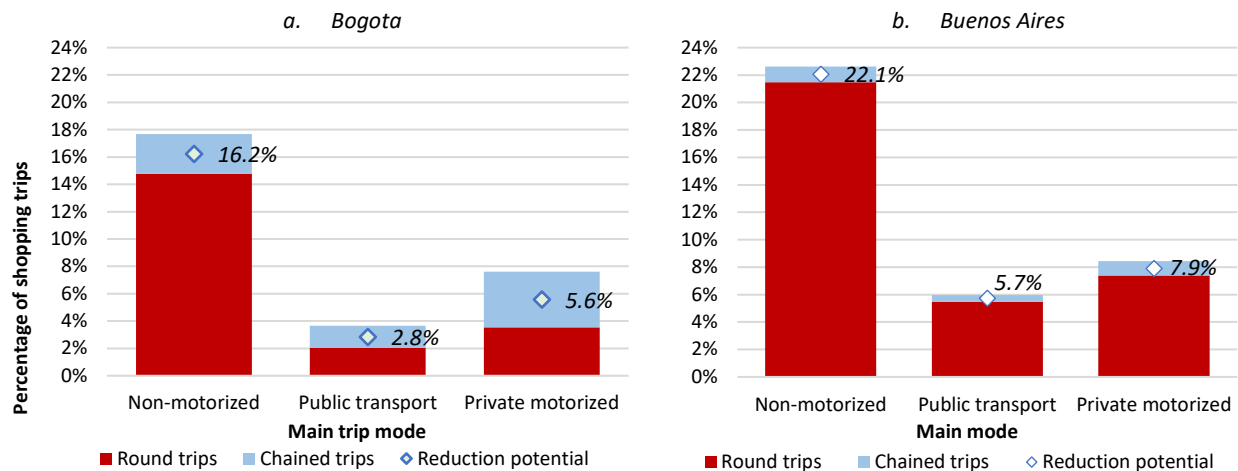


Source: Team's estimate based on EDM Bogota 2019 and ENMODO 2018

Shopping trips have a higher weight in short trips, reaching 26 percent for trips under 10 minutes in Bogota and 25 percent in Buenos Aires. Only 10 percent of trips that last 30-40 minutes and less than 5 percent among trips lasting over an hour in Bogota are shopping trips. In Buenos Aires, similarly, the weight of shopping trips falls rapidly with trip length, so that the potential for mobility reduction by e-commerce is above the overall average for trips under 20 minutes.

Existing studies have pointed out that, when assessing the potential traffic reduction effects of e-commerce, one has to consider that trips related to shopping are generally short distance and have a higher modal share of non-motorized modes in dense urban environments (Xi et al. 2020). Indeed, in both cities, trips for shopping purposes have a greater weight among non-motorized trips compared to trips made in other modes (Figure 3). In Bogota, shopping trips account for 18 percent of all non-motorized trips, while in Buenos Aires, the shopping trip share in non-motorized trips is between three and four times higher than in trips by other modes. In both cities, the share of linked trips over total trips for shopping purposes is lower among non-motorized trips. Shopping trips are more prevalent in private vehicle trips (8.4 percent in both cities) than in public transport trips. However, while in Buenos Aires only 12 percent of the private vehicle trips are chained, in the case of Bogota the share is over half, which limits the potential for trip reduction. As a result, the potential reduction of private vehicle trips due to e-commerce is about 5.6 percent in Bogota and 7.9 percent in Buenos Aires. Public transport is the mode with the lowest potential for trip reduction due to e-commerce (2.8 percent in Bogota and 5.7 percent in Buenos Aires).

Figure 3: Share of shopping trips, by type of travel chain and main mode of travel (%)



5. The impacts of the pandemic on e-commerce: Comparative results

The demand for e-commerce during the pandemic grew in both cities. In Bogota, the share of residents using the Internet to purchase products rose from 18.9 percent to 22.1 percent between 2019 and 2021, according to the Quality of Life Survey conducted annually by DANE. The survey data show a drop in e-commerce adoption in 2020 to 14.7 percent, a result that contradicts the general statistics of e-commerce growth in Colombia. This discrepancy may be related to a higher purchase frequency without expanding the number of users, to methodological differences in accounting for e-commerce transactions or to response biases in the surveys, among others. In Buenos Aires, the pandemic accelerated the growth in the number of online consumers, which had shown signs of stagnation in previous years; 20 percent of Internet users bought online for the first time during the quarantine. This growth was particularly strong in 2020, probably due to the limitations to make physical purchases as a result of mobility restrictions, with clothing and footwear, technological products and food products being the most commonly purchased types of goods (UAI 2020). According to the "Future Shopper Report 2021", Argentina was the LAC country with the highest e-commerce growth. In Argentina overall, the share of users choosing home delivery rose from 65 percent in 2019 to 80 percent in 2020, and in AMBA specifically the convenience associated with home delivery of products was the most mentioned reason for making online purchases (UAI 2020).

The typical profile of the online buyer in both cities is of high income. In Bogota, this is consistent with the purchase order data which shows a higher demand in locations in the north of the city. This trend strengthened during the pandemic, as e-commerce adoption in lower strata has stagnated. Already in 2019, more than 50 percent of people with households associated with stratum 6 used the Internet to shop online, compared to 6.3 percent among people in stratum 1 and 10.7 percent among people in stratum 2. The share of people buying online fell in 2020 to 1.1 percent among people with households associated with stratum 1 and to 5.8 percent among people with households associated with stratum 2. This could have been due to a reduction in consumption among these strata as a consequence of the pandemic crisis. The trend of stagnation of e-commerce among the lower strata and growth among the middle and upper strata was consolidated in 2021.

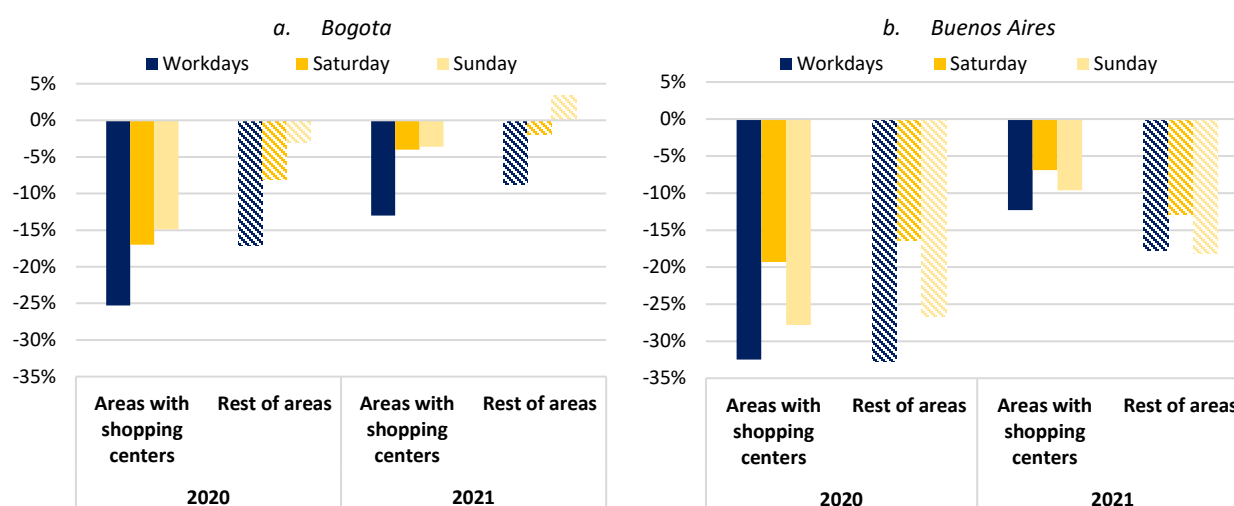
In 2019-2021, the share of Bogotá residents using the Internet to *sell* products rose from 1.3 percent to 3.5 percent. Although the adoption of this practice is still much lower than that of online shopping, its growth rate in the 2019-2021 period stands out, in which its adoption has practically tripled. The adoption of online product sales is also higher among the upper-middle socioeconomic strata, especially among strata 4 and 5. Unlike with online shopping, in this case the lower strata also registered growth in the adoption of product sales during

the pandemic, although lower than what was experienced by strata 4 and 5. In Buenos Aires, while the typical person buying online is of higher income, the pandemic narrowed the existing socioeconomic gap. Consumer surveys conducted by CACE indicate that consumers who started using e-commerce with the onset of the pandemic are mostly of lower-middle or lower socio-economic status. However, the size of the gap observed in 2018 was so high that it seems unlikely to have been completely reversed. Thus, if there is a substitution effect of shopping trips as a result of e-commerce, one would expect the drop in shopping trips to be larger among residents with higher socioeconomic status. Likewise, increases in urban distribution flows of goods would be expected to be higher for movements destined for high socioeconomic residential areas.

6. The changing urban freight dynamics during the pandemic and potential for trip elimination

In Bogota, total daily trips to areas associated with commercial activity – areas with shopping centers, areas near shopping centers, and areas that prior to the pandemic concentrated many shopping trips – fell more than trips to other areas in both 2020 and 2021 (Figure 4). This effect was particularly strong in the case of trips to areas with shopping centers: total trips to these areas in 2020 fell by 25.3 percent on weekdays compared to 2019 (compared to a 17.1-percent drop to other areas), by 17 percent on Saturdays (compared to 8.1 percent) and by 14.9 percent on Sundays (compared to 3 percent). The drop in mobility to commercial areas was particularly strong in the south of the city.

Figure 4: Change in trips to areas with shopping malls and the rest of areas in 2020 and 2021 vs. 2019 (%)



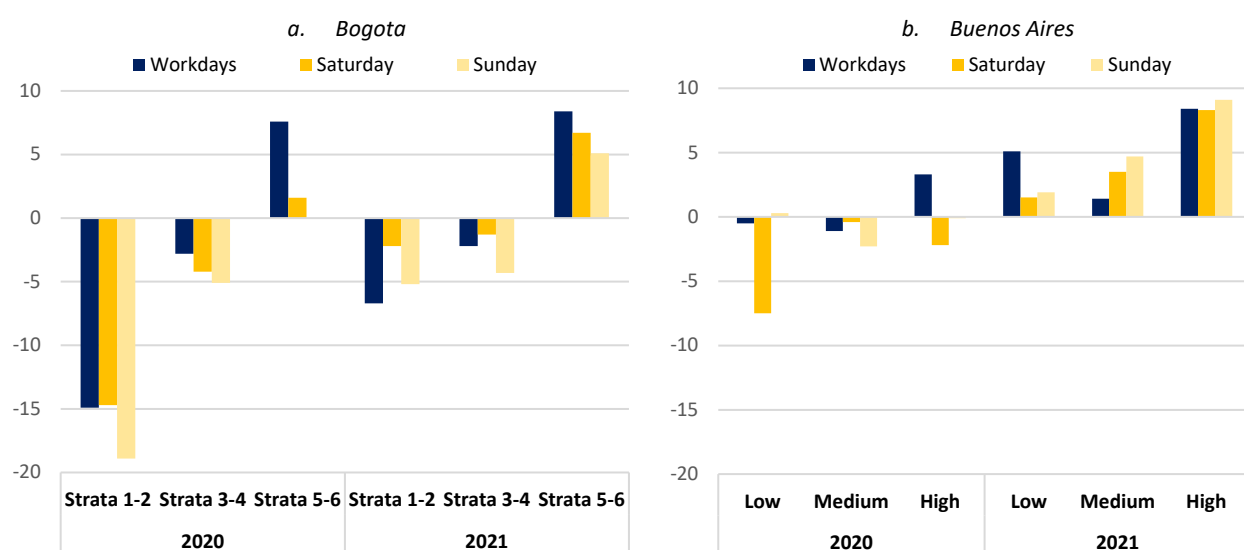
Source: Team's estimate based on CDR data

In Buenos Aires, total daily trips to areas associated with commercial activity in 2020 fell slightly more than in the city as a whole, although this was not the case of areas with shopping centers on weekdays, in which case the trips declined by 32.5 percent compared to 32.8 percent to the rest of areas. In 2021, the trend observed in 2020 was reversed: total daily trips to areas associated with commercial activity recovered faster than trips in the city overall. Of particular note are Sundays, when non-home-based trips with destination in areas with shopping malls was only 6.6 percent below the 2019 level, while non-home-based trips to other areas of the city were still 24.6 percent below.

Most of the other metrics suggest that in Bogota e-commerce did not reduce shopping related travel during the pandemic. The evolution of demand in times of the day that concentrate shopping travel according to the 2019 survey is contrary to what would be expected in case there was a substitution effect of trips through increased e-commerce adoption. The drop in travel to shopping areas was smaller in the 9:00-12:00 time slot – the window with most shopping-related mobility in 2019 – than in the day as a whole. Only in the case of the

19:00-20:00 time slot, which is the evening time with the greatest weight of shopping-purpose mobility in 2019, and only if looking at areas that have a shopping center, was there some evidence of travel behavior consistent with a substitution effect. Lower income travelers reduced mobility to shopping areas more than higher income groups, which is contrary to what would be expected if there were a relevant trip substitution effect as a consequence of e-commerce (Figure 5 (a)). In Buenos Aires, trips to shopping areas decreased more than to other areas of the city in 2020, and the decline was greater in the morning slot that previously concentrated shopping trips (9:00-12:00), suggesting a certain substitution effect. However, this trend reversed in 2021, suggesting that the steep drop in 2020 was largely a result of the enforcement of mobility restrictions during part of the pandemic period. Moreover, similarly to Bogota, in both 2020 and 2021, trips to shopping areas declined *less* from areas inhabited by people belonging to higher socioeconomic strata that are higher adopters of e-commerce (Figure 5 (b)).

Figure 5: Differences in change in travel to areas with shopping centers in 2020 and 2021 versus 2019, compared to other areas (percentage points), by traveler's socioeconomic strata/income



Note: Negative values indicate that trips to shopping center zones fell more or grew less than to other areas;

Source: Team's estimate based on CDR data

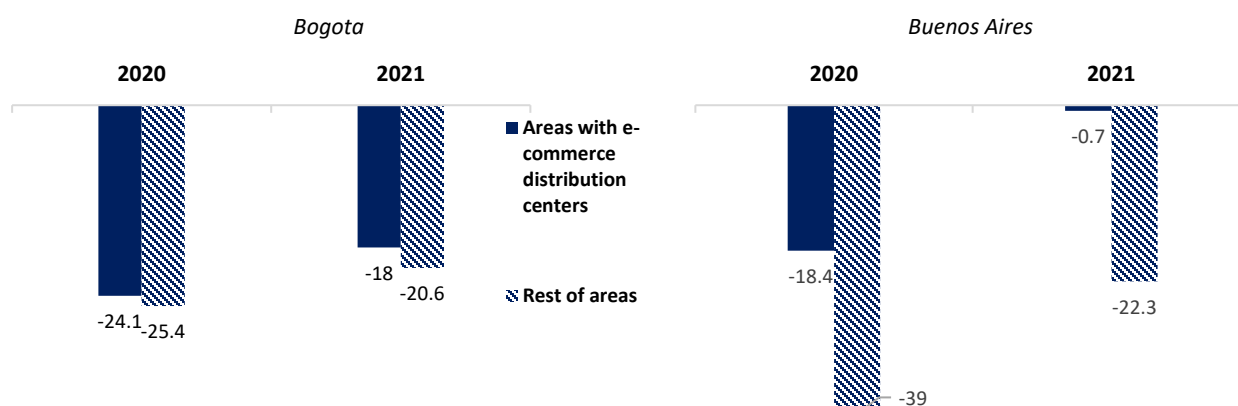
The results indicate that either there was no shopping trip demand substitution effect from increased e-commerce adoption or this effect was accompanied by other trends that affected mobility to shopping areas to a greater extent:

- A drop in employment in commercial areas, which would have caused the drop in mobility to these areas to be greater among the “compulsory” mobility segment than among the segments that include trips for shopping purposes.
- A reduction in consumption among the lower strata, which would have caused the drop in mobility to shopping areas to be greater among people of lower socioeconomic status, contrary to what would be expected as a result of the adoption of e-commerce.
- A complementarity effect of increase in telework, leading to an increase in shopping trips among the teleworking population and offsetting the possible substitution effect of e-commerce. Most studies on the effects of teleworking on travel demand have found this complementarity effect (e.g., Hook et al. 2020).

In Bogota and especially in Buenos Aires, carrier trips on weekdays originating in areas with e-commerce distribution centers fell less than those originating in other areas, consistent with the overall steep increase in e-commerce (Figure 6). In Bogota, this effect was accentuated in 2021 versus 2020, as the difference in the

decline was 1.3 percentage points in 2020 and 2.6 percentage points in 2021. The results for Saturday, and especially Sunday, point to an opposite trend. However, these results are considered less representative than the weekday results, judging by the distribution of e-commerce order shipments in Bogota by day of the week. Also in Buenos Aires, this effect was slightly more pronounced in 2021 compared to 2020, with the difference in the decline being 20.6 percentage points in 2020 and 21.6 percentage points in 2021. In 2021, weekday carrier trips originating in areas with logistics centers involved in e-commerce were only 0.7 percent lower than in 2019, compared with a 22.3 percent drop in all other areas. The effect is consistent with the increase in the proportion of consumers choosing home delivery, according to CACE reports. On Sundays the trend was the reverse, with carrier trips originating in areas with logistics centers falling more than in the city as a whole.

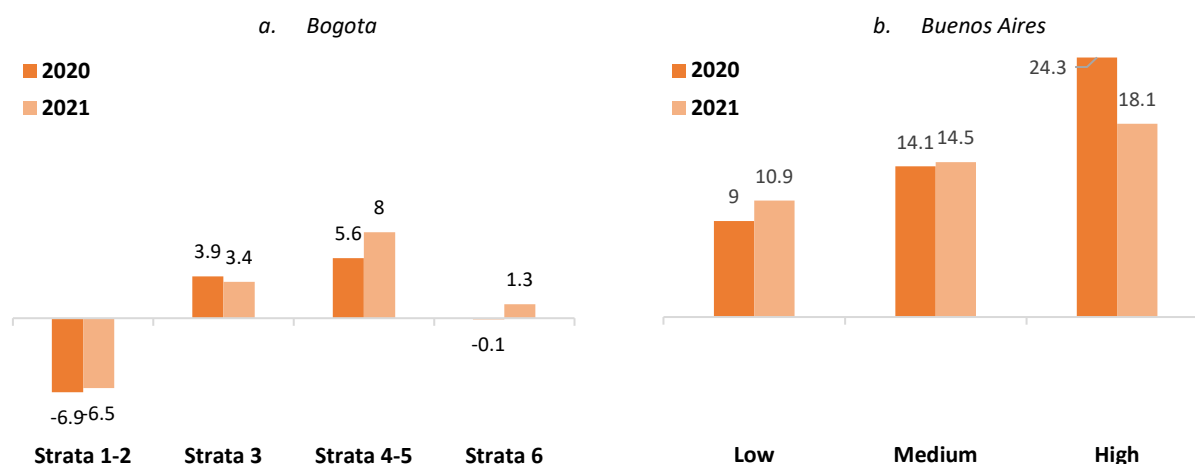
Figure 6: Change in the number of carrier trips from areas with e-commerce distribution centers, compared to change in the number of trips from all other areas, weekdays in 2020 and 2021 versus 2019 (%)



Source: Team's estimate based on CDR data

The number of trips by carriers originating in areas with e-commerce distribution centers and destined for residential areas with the highest e-commerce adoption – inhabited by higher income groups – fell the least in both cities (Figure 7). The trend holds for all days of the week and is especially significant on weekdays. This indicates that it is very likely that the growth of e-commerce implied an overall increase in freight traffic between 2019 and 2021.

Figure 7: Differences in change in carrier trips from areas with e-distribution centers in 2020 and 2021 versus 2019, compared to carrier trips from other areas (percentage points), by destination areas' socioeconomic strata/income



Source: Team's estimate based on CDR data

7. Policy implications

Analysis conducted as part of this study reveals that adoption of e-commerce could replace up to 9.1 percent of the trips made on weekdays in Bogota and between 7 percent and 12 percent of trips on weekdays in Buenos Aires. This corresponds to all trips for shopping purposes that are round trips and half of the trips for shopping purposes that are part of travel chains.

The reduction potential is greatest in the morning (though after peak hour), late afternoon, for short distance trips (< 10 minutes), and for trips made by non-motorized modes. The reduction potential among private vehicle trips is lower than among total trips, at 5.6 percent in Bogota and 8 percent in Buenos Aires. More than half of the trips by private vehicles for shopping purposes are part of travel chains, limiting their reduction potential. This implies that e-commerce has limits as a tool for reducing private vehicle use and externalities such as congestion during peak hours, given the concentration of shopping trips outside peak hours.

There is no evidence that the growth of e-commerce is having a significant substitution effect on shopping trips in either Bogota or Buenos Aires. Although the demand for trips to shopping areas in Bogota during 2019-2021 decreased more than to other areas of the city, the analysis by demand segments suggests that this was not due to the growth of e-commerce but that other phenomena (fall in employment in shopping areas, fall in consumption, etc.) are more explanatory of this evolution. Moreover, trips to shopping areas decreased less among the socioeconomic strata that are higher adopters of e-commerce. In Buenos Aires, the demand for trips to shopping areas decreased more than to other areas of the city in 2020, suggesting a certain substitution effect, but this trend reversed in 2021, suggesting that the steeper drop in 2020 was strongly influenced by the enforcement of mobility restrictions. Moreover, consumer surveys by CACE indicate that among online shoppers there was an *increase* in visits to physical stores.⁶ In other words, e-commerce currently generates more traffic than it avoids, as a consequence of an increased number of freight vehicle trips and a very limited substitution effect on the demand for private vehicle trips. This means that for short- and medium-term sustainable mobility strategies (e.g., mobility master plans), e-commerce is an additional challenge to manage and not a significant opportunity that can help solve mobility problems.

Despite the limitations of e-commerce as an effective tool in reducing congestion, it opens up another set of opportunities for mobility planners. If urban distribution fleets are electrified faster than the private passenger vehicle fleet, e-commerce can be effective at reducing emissions from the transport sector in a future scenario of mass adoption. In this scenario, any substitution effect would have a positive impact on emissions reduction by replacing some private vehicle trips with electric freight vehicle trips. The impact of e-commerce on road accidents and the cost of road maintenance (elements highlighted by the Bogota Mobility Master Plan as impacts of urban freight distribution) will be more complex to address, as it depends solely on the optimization of distribution chains and routes to avoid a disproportionate increase in freight vehicle kilometers traveled as a result of the growth of e-commerce. This optimization can also contribute to reducing emissions without waiting for a complete electrification of distribution fleets.

The associated recommendations for policy are as follows:

- ***Establish close relationships with e-commerce stakeholders.*** In a context of limited legal capacity to prescribe data sharing by the private sector, transport authorities should invest more resources in establishing close relationships with local players in the e-commerce sector. This may lead to a greater willingness to share information and data on e-commerce demand and its implications for mobility (Rojas-Huerfano et al. 2018).

⁶ The rate of returns of goods in the City of Buenos Aires is around 3 percent (Abad et al. 2022), which generates further need for physical interaction and travel.

- ***Treat households as freight attractor nodes in mobility surveys.*** Freight transport studies usually focus on studying trip generation and attraction by production, logistics and sales nodes, including only retail as the final destination point and without considering households as a point of attraction for freight trips. Household mobility surveys can include basic questions about online shopping habits without significantly increasing the response time.
- ***Manage e-commerce as a mobility-generating challenge in the short-to-medium term.*** The existing pressure on mobility planning to find ways to reduce externalities in the transport sector means that “virtualization” of activities is routinely seen as a powerful tool to reduce mobility demand (Pettersson et al. 2018). In the case of e-commerce, international evidence and the results of the analysis conducted this study suggest that this is not the case; as of today there is no substitution effect of shopping trips that compensates for the increase in freight vehicle traffic. In this sense, e-commerce differs from the effects of other virtualization processes (telework, online leisure, etc.) in which there may be more opportunity to reduce mobility demand without generating other types of trips. In any case, the effect of each social trend on mobility demand has to be carefully analyzed prior to pinning hopes on it in terms of mobility reduction. For example, the available evidence on the effects of teleworking suggests complementarity effects with other travel purposes and increases in distances traveled (Ravalet and Rerat 2019).
- ***Focus on the optimization of distribution routes and the electrification of fleets.*** The optimization of distribution routes is the measure that can most effectively mitigate the impacts of e-commerce on mobility, as it can reduce the number of kilometers traveled by freight vehicles in the city, with all that it entails in terms of reducing emissions, mitigating road accidents associated with this type of vehicles and containing congestion. The co-creation of more sustainable alternatives for distribution together with the sector stakeholders must be a priority for the transport authorities. This optimization includes the implementation of freight consolidation nodes and alternative delivery mechanisms (e.g., automatic ticketing machines at transport interchanges). After the optimization of distribution routes, fleet electrification is the second measure with the highest return, as it has a direct impact on emissions from e-commerce-generated freight vehicle traffic. Transport authorities can incentivize this electrification by segmenting traffic requirements and restrictions according to vehicle profile. Finally, the trend of reducing order size and delivery time increases the generation of freight vehicle trips associated with e-commerce. Potential measures to reduce the number of freight vehicle miles traveled include, among others, urban consolidation centers, proximity locker delivery, etc.

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